

# Harshwardhan Patil

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## Summary

Full-Stack Engineer with **5+ years of experience** building scalable, distributed systems and high-performance APIs across frontend, backend, and cloud. Skilled in **system design**, CI/CD on AWS, production monitoring, and delivering **fault-tolerant, data-consistent services** at scale in **Agile** environments.

## Technical Skills

- **Languages & Frameworks:** Java, TypeScript, JavaScript, SQL, Spring Boot, Angular, Node.js, Python, FastAPI
- **Cloud & DevOps:** AWS (EC2, ECS, S3, RDS, CloudFront, ALB, Secrets Manager, Lambda), Docker, Jenkins, Git, GitHub Actions, Nginx, Cloudflare
- **Databases:** PostgreSQL, Redis, MySQL, Neo4j, Elasticsearch, MongoDB, Schema design, Database indexing, Query optimization, Caching strategies
- **Testing & Security:** JUnit, Mockito, Jasmine, Vitest, ESLint, SonarQube, Postman, JWT, RBAC, HTTPS/TLS, CORS, CSRF/XSS protection
- **Design & Architecture:** System Design, Design Patterns, SOLID principles, Distributed backend services, Concurrency handling, Session management
- **AI & ML:** Spring AI, OpenAI API, Prompt Engineering, Embeddings, Vector Search, RAG Pipelines, pgvector

## Professional Experience

- Software Engineer, Clarivate Analytics** – Bangalore, India Aug 2023 - Jul 2024
- Developed scalable Spring Boot REST APIs with DTO contracts and standardized error handling, reducing breaking changes by **30%** across releases.
  - Implemented **JWT**-based multi-session management across backend services, capping concurrent logins at 12 per user with token invalidation on breach.
  - Optimized **PostgreSQL** schemas with composite indexing across reporting and search workloads for drug and patent datasets, cutting full scans.
  - Containerized and deployed microservices to **EC2** using Docker and Jenkins **CI/CD** pipelines, supporting **50+ production releases** with **zero rollbacks**.
  - Enforced role-based access control across backend and frontend layers, applying tiered and region-based data visibility for **10+ access variants**.
  - Implemented low-latency **Elasticsearch**-powered typeahead and autocomplete, improving search response times by **40%** over large-scale datasets.
  - Developed dynamic PPT and PDF export pipelines using **PptxGenJS** and **jsPDF**, generating **100+ customized reports** with user-selected filters.
  - Optimized **PostgreSQL** query logic for PPT export, decoupling bulk multi-filter chart fetching from single-chart UI rendering to reduce DB round-trips.
  - Achieved **90%** test coverage using **JUnit** and **Mockito** for backend and **Jasmine** for Angular, covering unit, branch, and line coverage across all modules.

- Associate Software Engineer, Clarivate Analytics** – Bangalore, India Aug 2021 - Jul 2023
- Integrated **Elasticsearch** into Spring Boot, enabling advanced full-text queries across **millions of drug records** and reducing search latency by **30%**.
  - Migrated **5+ core business flows** from Angular 8 to **Angular 13**, rebuilding components with a modular architecture and improved state management.
  - Built secure request-handling across backend services using **Spring Security**, **CORS/CSRF**, and input validation to guard against unauthorized access.
  - Improved responsiveness of data-heavy chart and table views by **20–25%** through lazy loading, **RxJS** streams, and Angular lifecycle optimizations.
  - Reduced page load times by **20%** by applying component-level **OnPush** change detection and optimizing data-fetching patterns across data-heavy views.
  - Built **D3.js** chart components with grouping and filtering aggregations for **Angular**, powering **15+ analytics views** across drug intelligence dashboards.
  - Architected a configurable **Angular** table framework with sorting, pagination, and filters; adopted across **3 internal teams**, cutting duplication by **50%**.

## Education

**University of Texas at Arlington**    *Master's in Software Engineering*    *GPA: 3.6 / 4.0*    Aug 2024 - May 2026

## Projects

- TinyURL — URL Shortener**    🔗 🔄 Feb 2026 - Mar 2026
- Production-grade URL shortening service with QR code generation, link expiry, and a fully automated cloud deployment pipeline.*
- Tools:** Spring Boot, Angular, PostgreSQL, AWS (EC2, RDS, S3, CloudFront, ALB), Docker, Cloudflare, Nginx, Lambda, EventBridge, SQS DLQ, CloudWatch
- Designed **6-layer request path** across Cloudflare, CloudFront, ALB, Nginx, Spring Boot, and PostgreSQL, separating SPA delivery from API and redirects.
  - Hardened API traffic with **Nginx rate limits**, **Bucket4j token buckets**, **layer-7** bot/scanner filters, per-IP size caps, and strict **HSTS/CSP** secure headers.
  - Designed serverless EC2/RDS scheduler with Lambda, EventBridge, CloudWatch alarms, SNS, and SQS DLQ, reducing monthly cloud cost by **~40%**.
  - Built zero-SSH **CI/CD** with GitHub Actions, AWS OIDC, SSM RunCommand, Docker smoke tests, cosign signing, and auto CloudFront cache invalidation.

- diagram-sync — Diagram-as-Code CLI**    🔄 📄 ⬇ 3,323 May 2026 - Present
- Published npm package that renders architecture diagrams from 7 source formats into versioned SVG/PNG outputs via `npa diagram-sync`.*
- Tools:** TypeScript, Node.js, Vitest, ESLint, GitHub Actions
- Engineered **provider-registry architecture** mapping 14+ file extensions to isolated rendering adapters with runtime graceful degradation on missing tools.
  - Developed recursive repo scan, `--files`, and `--changed` via `git diff HEAD` with untracked support, enabling CI to regenerate only PR-changed files.
  - Architected **tiered format resolution** across CLI flag, job, and global config with per-provider validation and actionable errors for unsupported formats.

- Employee Management System**    🔗 🔄 Dec 2024 - May 2025
- Enterprise-grade system for managing employees, teams, projects, and access control with production-grade deployment and observability.*
- Tools:** Spring Boot, Angular 19, PostgreSQL, Redis, AWS EC2, SSM, Docker, Docker Compose, Nginx, Cloudflare, GitHub Actions
- Engineered a role-aware **JWT-based RBAC** backend across employees, departments, projects, tasks, and locations with DTO/service/repository layers.
  - Secured production traffic with **Cloudflare-proxied DNS**, Nginx gateway routing, SSL/TLS, BCrypt auth, health checks & containerized EC2 deployment.
  - Built **REST APIs** with pagination, filtering, search, and dynamic dashboard metrics; added reusable **Angular Material** table views with shared components.

- RAG Document Engine**    🔄 Jun 2026 - Present
- End-to-end RAG system with multi-format ingestion, hybrid retrieval, LLM answer generation, and role-based document access control.*
- Tools:** Python, NumPy, OpenAI API, ChromaDB, sentence-transformers
- Built **multi-format ingestion** (PDF, Word, Markdown) with overlapping chunking, deduplication, and **1536-dim** L2-normalized embeddings into ChromaDB.
  - Engineered **hybrid retrieval** via ensemble sparse BM25 + cross-encoder reranking with **tiktoken** token budgeting for citation-grounded LLM generation.
  - Implemented **multi-turn chat** with history truncation and **RBAC-scoped metadata filtering** in ChromaDB enforcing per-user document access at retrieval.